Colégio Agostiniano Mendel

Componente Curricular: Física – Dinâmica
Professor(a): Rosana C. Pádua **Série:** 2º **Data:** Agosto
Conteúdo: Quantidade de movimento - Impulso - Teorema do Impulso

1. $v_0 = 0$ → $\vec{F_m}$ → $\vec{v}$ (+→)

a) $Q = m \cdot v \Rightarrow Q = 0{,}4 \cdot 24 \Rightarrow Q = 9{,}6\ kg \cdot m/s$

b) $I_R = \Delta Q \Rightarrow F_m \cdot \Delta t = mv - m\vec{v_0}^{\,0} \Rightarrow F_m \cdot 3 \cdot 10^{-2} = 9{,}6 \Rightarrow F_m = 320\ N$

2. $\vec{v_0}$ | $\vec{v}$ (+←)

antes | depois

$\vec{F_m}$

$I_R = \Delta Q \Rightarrow F_M \Delta t = mv - mv_0$

$F_M \cdot 0{,}15 = 1500 \cdot 3 - 1500(-15)$

$F_M \cdot 0{,}15 = 4500 + 22500 \Rightarrow 0{,}15 F_M = 27000$

$F_M = 180000\ N$

3. $\vec{v_0}$ ← $\vec{I}$ | $v = 0$; $\vec{F}$ (+←)

$I_R = \Delta Q$

$F_R \cdot \Delta t = m\vec{v}^{\,0} - mv_0$

① $10 \cdot 20 = M_1(-v_0)$ ② $20 \cdot 20 = M_2(-v_0)$ ① / ② $\Rightarrow \frac{-M_1 v_0}{-M_2 v_0} = \frac{200}{400}$

$\frac{M_1}{M_2} = \frac{1}{2}$

4. F(N): 60, 40, 20; t(s): 0, 2, 4, 12, 16; áreas 1, 2, 3

$I_R = \Delta Q \Rightarrow A = mv - mv_0$

$\left(\frac{40+20}{2}\right) \cdot 2 + 8 \cdot 40 + \left(\frac{60+40}{2}\right) \cdot 4 = 58v - 58 \cdot 3$

$60 + 320 + 200 = 58v + 174$

$580 = 58v - 174 \Rightarrow 58v = 754 \Rightarrow v = 13\ m/s$

5. F(N): 20, F_1; t(s): 0, 1, 3, 7, 9

$I_R = \Delta Q$

a) $I_R = F_1 \cdot \Delta t$

$30 = F_1 \cdot 2$

$F_1 = 15\ N$

b) $I_R = \Delta Q$

$A = \Delta Q$

$\left(\frac{20+15}{2}\right) \cdot 4 = \Delta Q \Rightarrow \Delta Q = 70\ N.s$

6. $\vec{Q}$ ↑ ; $\vec{Q_0}$ →

$\star\ \vec{I_R} = \Delta\vec{Q}$ $\star\ \Delta\vec{Q} = \vec{Q} - \vec{Q_0}$

$\Delta\vec{Q} = \vec{Q} + (-\vec{Q_0})$

$\vec{Q}$, $-\vec{Q_0}$, $\Delta\vec{Q}$ ($\vec{I_R}$)

$Q = Q_0 = mv$

$\therefore |\Delta Q = \sqrt{2}\, mv|$

Diagonal do quadrado

Opção p/ a fig

$-\vec{Q_0}$, $\Delta\vec{Q} = \vec{I_R}$, $\vec{Q}$

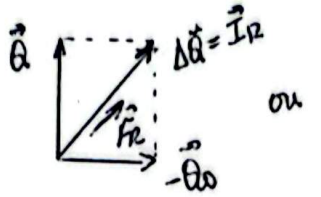
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7. $Q = mv$ $\begin{cases} \cdot Q_0 = 1.3 = 3\ kg\cdot m/s \\ \cdot Q = 1.4 = 4\ kg\cdot m/s \end{cases}$

$\vec{Q}$ ↑ ← $\vec{Q_0}$

$I_R = \Delta\vec{Q}$ ($I_R = \Delta Q$)

$\Delta\vec{Q} = \vec{Q} + (-\vec{Q_0})$



ou



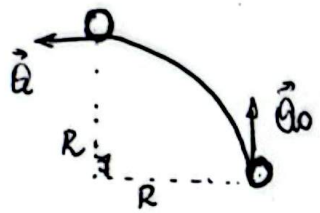
$\Delta Q = \sqrt{3^2+4^2}$

$\Delta Q = 5\ kg\cdot m/s$

$\therefore I_R = 5\ N.s$

$I_R = F_R \cdot \Delta t \Rightarrow 5 = F_R, 0{,}5 \Rightarrow F_R = 10\ N$

8.

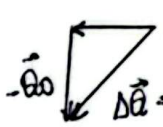


$Q = Q_0 = m \cdot v_0$

$Q_0 = 0{,}1 \cdot 2$

$Q_0 = 0{,}2\ kg\cdot m/s$ ($Q = 0{,}2\ kg\cdot m/s$)

$\Delta\vec{Q} = \vec{Q} + (-\vec{Q_0})$



$\Delta Q = 0{,}2\sqrt{2}\ kg\cdot m/s$

$\cdot I_R = \Delta Q \Rightarrow I_R = 0{,}2\sqrt{2}\ N.s$

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